

## Experiment No. 25

Write a Prolog Program for fruit and its color using Back Tracking.

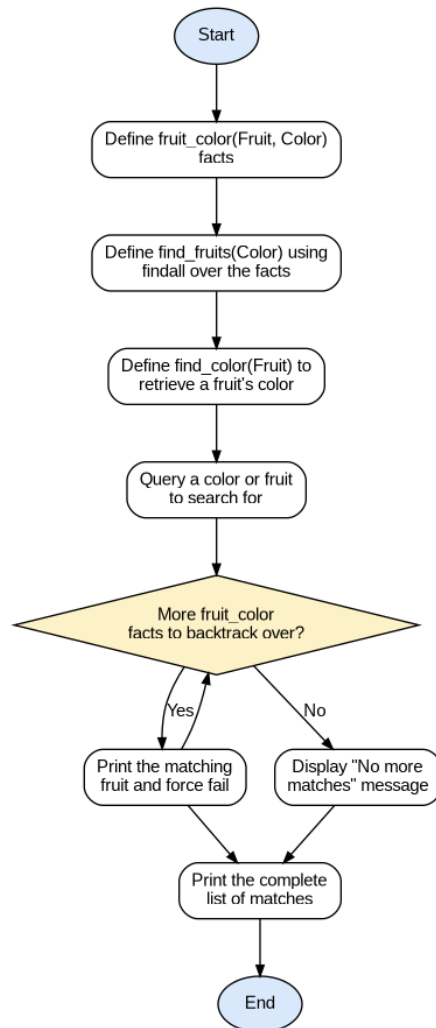
### Aim

To write a Prolog program that stores facts about various fruits and their colors, and uses backtracking to retrieve all matching fruits for a given color and to display each match in turn.

### Algorithm

1. Start.
2. Define facts `fruit_color(Fruit, Color)` representing the color of each fruit.
3. Define the rule `find_fruits(Color)` that uses `findall` to collect all fruits matching a given color.
4. Define the rule `find_color(Fruit)` that retrieves the color of a specific fruit by matching the facts.
5. Define the rule `show_all_red` that backtracks over the `fruit_color` facts for red fruits one at a time using fail-driven backtracking.
6. In `show_all_red`, print each red fruit found and then force failure using `fail` to trigger backtracking to the next matching fact.
7. Define a second clause of `show_all_red` that succeeds after backtracking is exhausted, and display a completion message.
8. Query `find_fruits` for different colors and display the resulting lists.
9. Query `find_color` for a specific fruit and display its color.
10. Query `show_all_red` to demonstrate backtracking through all matching facts.
11. Stop.

### Flowchart



### Program

```

fruit_color(apple, red).
fruit_color(banana, yellow).
fruit_color(grape, green).
fruit_color(cherry, red).
fruit_color(mango, yellow).
fruit_color(lime, green).
fruit_color(strawberry, red).

```

find\_fruits(Color) :-

```

    findall(Fruit, fruit_color(Fruit, Color), Fruits),
    format("Fruits with color ~w: ~w~n", [Color, Fruits]).

```

find\_color(Fruit) :-

```

    fruit_color(Fruit, Color),
    format("~w is ~w in color~n", [Fruit, Color]).

```

```
show_all_red :-  
    format("Backtracking through all red fruits:~n"),  
    fruit_color(Fruit, red),  
    format(" - ~w~n", [Fruit]),  
    fail.  
show_all_red :-  
    format("No more red fruits.~n").
```

```
main :-  
    find_fruits(red),  
    find_fruits(yellow),  
    find_fruits(green),  
    find_color(apple),  
    show_all_red.
```

#### Result

The program was executed successfully and the output was verified as correct.

?- main.

Fruits with color red: [apple,cherry,strawberry]

Fruits with color yellow: [banana,mango]

Fruits with color green: [grape,lime]

apple is red in color

Backtracking through all red fruits:

- apple
- cherry
- strawberry

No more red fruits.

true.

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

fruitcolor.pl - AI-Lab - Visual Studio Code

EXPLORER

AI-Lab

sum\_n.pl

name\_dob.pl

student\_teacher.pl

planets.pl

hanoi.pl

birdflight.pl

familytree.pl

dietsystem.pl

monkeybanana.pl

fruitcolor.pl

fruitcolor.pl

1 fruit\_color(apple, red).

2 fruit\_color(banana, yellow).

3 fruit\_color(grape, green).

4 fruit\_color(cherry, red).

5 fruit\_color(mango, yellow).

6 fruit\_color(lime, green).

7 fruit\_color(strawberry, red).

8

9 find\_fruits(Color) :-

10 findall(Fruit, fruit\_color(Fruit, Color), Fruits),

11 format("Fruits with color ~w: ~w~n", [Color, Fruits]).

12

13 find\_color(Fruit) :-

14 fruit\_color(Fruit, Color),

15 format("~w is ~w in color~n", [Fruit, Color]).

16

17 show\_all\_red :-

18 format("Backtracking through all red fruits:-~n"),

19 fruit\_color(Fruit, red),

20 format(" ~w~n", [Fruit]),

21 fail.

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS C:\Users\Athavan\AI-Lab> swipl fruitcolor.pl

Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.

Fruits with color red: [apple,cherry,strawberry]

Fruits with color yellow: [banana,mango]

Fruits with color green: [grape,lime]

apple is red in color

Backtracking through all red fruits:

~ apple

~ cherry

~ strawberry

No more red fruits.

true.

?-

Prolog

UTF-8 LF Spaces: 4